Nombre del juego: DH Endless Runner Duck

Versión 1.0

Autor: Diego Francisco Hernández Ramírez

Motor y versión: Godot Engine 4.2.2

Lenguaje: GDScript

Fecha: 30 de abril del 2026

1. Resumen ejecutivo

QUÉ ES EL PROYECTO.

DH Endless Runner Duck es un videojuego 2D del género endless runner desarrollado con Godot Engine 4.2.2.
El proyecto fue concebido como un producto completo multiplataforma (PC y Android), incorporando progresión por niveles, sistema de vidas, guardado persistente, desbloqueo de mundos y adaptación de controles táctiles.

2. Objetivos del proyecto

OBJETIVO GENERAL

Diseñar y desarrollar un videojuego 2D completo que integrara:

- gameplay sólido
- arquitectura modular
- persistencia de datos
- UI funcional
- exportación multiplataforma

OBJETIVOS TÉCNICOS

- implementar sistema de guardado JSON
- crear managers globales
- diseñar escenas reutilizables
- portar a Android
- adaptar input touch

3. Concepto del juego

GÉNERO

Arcade / Endless Runner

MECÁNICA PRINCIPAL

Jugador:

🦆 pato

Objetivo:

esquivar obstáculos y sobrevivir.

Movimiento:

3 carriles.

CORE LOOP

Iniciar partida
↓
Esquivar obstáculos
↓
Aumentar distancia
↓
Subir nivel
↓
Incrementar dificultad
↓
Desbloquear mundos
↓
Repetir

4. Sistemas implementados

SISTEMA DE MOVIMIENTO

- lane switching
- interpolación suave
- swipe mobile
- teclado PC

SISTEMA DE VIDAS

- 3 vidas
- invulnerabilidad temporal
- feedback visual
- hit animation

SISTEMA DE PROGRESO

- distancia
- niveles
- escalada dificultad

SAVE SYSTEM

Persistencia:

```
{
  "coins": 0,
  "high_score": 0,
  "unlocked_worlds": [true,false,false]
}
```

WORLD UNLOCK SYSTEM

🌱 LAND
↓
🌊 RIVER
↓
☁️ SKY

5. Arquitectura del software

Diagrama:

```
World
 ├ Player
 ├ ObstacleSpawner
 ├ AudioManager
 ├ UI
 └ Managers

Autoload:
 ├ SaveManager
 ├ LevelManager
 └ WorldManager
```

SAVEMANAGER

Encargado de:

- guardar
- cargar
- persistencia

LEVELMANAGER

Encargado de:

- distancia
- niveles
- multiplicadores

WORLDMANAGER

Encargado de:

- cambio de mundos
- señales globales

6. Adaptación móvil

PROBLEMAS RESUELTOS

- orientación
- touch input
- swipe movement
- pausa touch
- export APK firmado

SOLUCIÓN IMPLEMENTADA

Swipe horizontal:

izquierda / derecha.

Botón pausa touch.

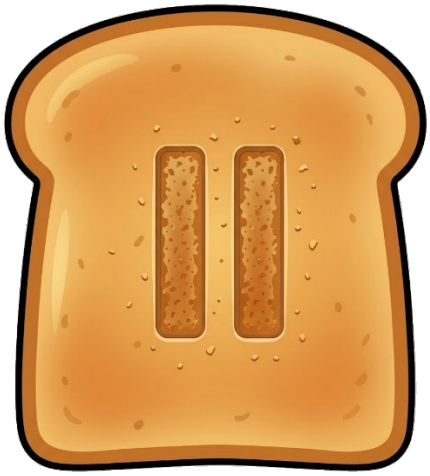
7. Arte y UI

DH Endless Runner Duck (DEBUG)
DH Endless Runner Duck
Play
Worlds
Exit
V. 1.0

Play

Worlds

Exit



8. Capturas

DH Endless Runner Duck (DEBUG)
DH Endless
Runner Duck
Play
Worlds
Exit
V. 1.0

DH Endless Runner Duck (DEBUG)
9
7890 m

DH Endless Runner Duck (DEBUG)
0
65 m
PAUSE
RESUME

DH Endless Runner Duck (DEBUG)
0
65 m
PAUSE
RESUME

DH Endless Runner Duck (DEBUG)
5
4519 m
Nivel Completado
Next World
Continue World Infinite

DH Endless Runner Duck (DEBUG)
10
8059 m
¡El pato chocó!
Distancia recorrida: 8059 m
REINICIAR

DH Endless Runner Duck (DEBUG)
9
7890 m

9. Retos técnicos

SISTEMA DE PAUSA

PROBLEMA:

pausar congelaba UI.

SOLUCIÓN:

PauseMenu scene independiente.

Android Signing

PROBLEMA:

APK inválido.

SOLUCIÓN:

keystore release.

Touch Input

PROBLEMA:

swipe detectado incorrectamente.

SOLUCIÓN:

tracking correcto de touch_start.

10. Tecnologías utilizadas

Tecnología	Uso
Godot Engine	Motor
GDScript	Lógica
Json	Save
Git	Control versiones
GitHub	Repositorio
Itch.io	Distribución

11. Conclusiones

El proyecto permitió consolidar habilidades de programación orientada a videojuegos, diseño modular, manejo de estados, persistencia de datos y despliegue multiplataforma.

12. Roadmap

v1.1:

- shop
- skins
- achievements
- ads reward

v2:

- online leaderboard
- analytics

1. Executive summary

WHAT IS THE PROJECT.

DH Endless Runner Duck is a 2D video game of the endless runner genre developed with Godot Engine 4.2.2.

The project was conceived as a complete cross-platform product (PC and Android), incorporating level progression, life system, persistent saving, unlocking worlds and adapting touch controls.

2. Project objectives

GENERAL OBJECTIVE

Design and develop a complete 2D video game that integrates:

- solid gameplay
- modular architecture
- data persistence
- Functional UI
- cross-platform export

TECHNICAL OBJECTIVES

- implement JSON save system
- create global managers
- design reusable scenes
- port to Android
- adapt input touch

3. Game concept

GENDER

Arcade /Endless Runner

MAIN MECHANICS

Player:

🦆 duck

Objective:

avoid obstacles and survive.

Movement:

3 lanes.

CORE LOOP

Start game

↓

Avoid obstacles

↓

Increase distance

↓

Level up

↓

Increase difficulty

↓

Unlock worlds

↓

Repeat

4. Implemented systems

MOVEMENT SYSTEM

• lane switching

• smooth interpolation

• swipe mobile

• PC keyboard

SYSTEM OF LIVES

• 3 lives

• temporary invulnerability

• visual feedback

• hit animation

PROGRESS SYSTEM

- distance
- levels
- difficulty climbing

SAVE SYSTEM

Persistence: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{
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}
```

WORLD UNLOCK SYSTEM

🌱 LAND

↓

🌊 RIVER

↓

☁️ SKY

5. Software architecture

DIAGRAM:

World

├ Player

├ ObstacleSpawner

├ AudioManager

├ U.I.

└ Managers

AUTOLOAD:

├ SaveManager

├ LevelManager

└ WorldManager

SAVEMANAGER

In charge of:

• save

• charge

• persistence

LEVELMANAGER

In charge of:

• distance

• levels

• multipliers

WORLDMANAGER

In charge of:

- change of worlds
- global signals

6. Mobile adaptation

SOLVED PROBLEMS

• orientation

• touch input

• swipe movement

• touch pause

• signed APK export

SOLUTION IMPLEMENTED

Horizontal swipe:

left right.

Pause touch button.

7. Arte y UI

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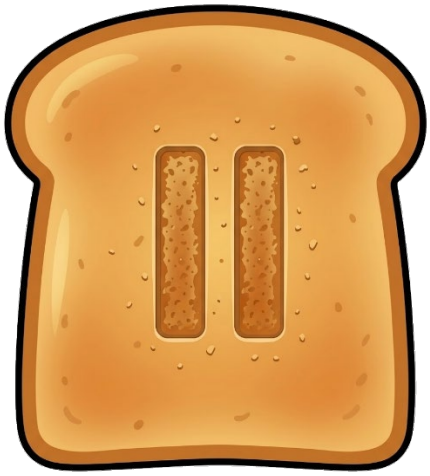

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7890 m

9. Technical challenges

PAUSE SYSTEM

Problem:

pausing froze UI.

Solution:

PauseMenu independent scene.

ANDROID SIGNING

PROBLEM:

Invalid APK.

SOLUTION:

keystore release.

TOUCH INPUT

PROBLEM:

swipe detected incorrectly.

SOLUTION:

correct touch_start tracking.

10. Technologies used

Technology Use

Godot Engine Engine

GDScript Logic

Json Save

Git Control versions

GitHub Repository

Itch.io Distribution

11. Conclusions

The project allowed the consolidation of video game-oriented programming skills, modular design, state management, data persistence and multiplatform deployment.

12. Roadmap

v1.1:

• shop

• skins

• achievements

• ads reward

v2:

• online leaderboard

• analytics